

EXERCISE 1.

1.1 (2 pt)

$$V = [H^1(\Omega)]^2 \quad Q = L^2(\Omega)$$

$\forall t \in (0, T)$, Find $\vec{u}(t) \in V$, $p(t) \in Q$:

$$\begin{cases} \int_{\Omega} \frac{\partial u}{\partial t} v + a(u, v) + b(v, p) + \int_{\Omega} u \cdot \nabla u v = \int_{\Omega} f v + \int_{\partial \Omega} \psi v \quad \forall v \in V \\ b(u, q) = 0 \quad \forall q \in Q \\ u|_{t=0} = 0 \quad x \in \Omega \end{cases} \quad a(u, v) = \int_{\Omega} v \nabla u \nabla v \quad b(v, q) = - \int_{\Omega} q \operatorname{div} v$$

1.2 (1 pt)

$\forall t \in (0, T)$, Find $u_h(t) \in V_h$, $p_h(t) \in Q_h$:

$$\begin{cases} \int_{\Omega} \frac{\partial u_h}{\partial t} v_h + a(u_h, v_h) + b(v_h, p_h) + \int_{\Omega} u_h \cdot \nabla u_h v_h = \int_{\Omega} f_h v_h + \int_{\partial \Omega} \psi_h v_h \\ b(u_h, q_h) = 0 \quad \forall q_h \in Q_h \\ u_h|_{t=0} = 0 \quad x \in \Omega \end{cases} \quad \forall v_h \in V_h$$

1.3 (2 pt)

$\forall k = 0, 1, \dots$

$$\begin{cases} \int_{\Omega} \frac{1}{\Delta t} (u_h^{k+1} - u_h^k) v_h + a(u_h^{k+1}, v_h) + b(v_h, p_h^{k+1}) \\ + \int_{\Omega} u_h^k \cdot \nabla u_h^k v_h = \int_{\Omega} f^{k+1} v_h + \int_{\partial \Omega} \psi^{k+1} v_h \quad \forall v_h \in V_h \\ b(u_h^{k+1}, q_h) = 0 \quad \forall q_h \in Q_h \end{cases}$$

$$u_h^0 = u_{0h} \in V_h$$

The couple (u_h^{k+1}, p_h^{k+1}) is the solution, at any k , of a generalized Stokes problem. The latter is well-posed since the couple (V_h, Q_h) is the Taylor-Hood couple that satisfies the LBB condition.

1.4 (3 pt) stability condition:

$$\Delta t \leq C \frac{h}{\max |u^k|} \quad \text{at any } k$$

convergence result (error estimate):

$\forall$ time t^k ,

$$\begin{aligned} & \|u(t^k) - u_h^k\|_{H^1(\Omega)} + \|p(t^k) - p_h^k\|_{L^2(\Omega)} \\ & \leq C(\Delta t + h^2), \quad C \text{ depending on } u, p. \end{aligned}$$

EXERCISE 2

2.1 (1 pt)

Find $u \in V = H^1(\Omega)$ s.t. $a(u, v) = \int_{\Omega} f v \quad \forall v \in V$

$$\text{with } a(u, v) = \int_{\Omega} \alpha \nabla u \nabla v - \int_{\Omega} \vec{\beta} \cdot \nabla u v + \int_{\Omega} \gamma u v$$

2.2 (3 pt)

Lax-Milgram Lemma hypotheses

• continuity of $a(\cdot, \cdot)$:

$$\begin{aligned} |a(u, v)| & \leq \alpha \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + |\vec{\beta}| \|\nabla u\|_{L^2} \|v\|_{L^2} + \gamma \|u\|_{L^2} \|v\|_{L^2} \\ & \leq \underbrace{\max(\alpha, |\vec{\beta}|, \gamma)}_M \|u\|_{H^1} \|v\|_{H^1} \end{aligned}$$

• coercivity of $a(\cdot, \cdot)$

$$a(v, v) = \alpha \|\nabla v\|_{L^2}^2 - \underbrace{\int_{\Omega} \vec{\beta} \cdot \frac{1}{2} \nabla(v^2)}_{\frac{1}{2} \operatorname{div} \vec{\beta} \int_{\Omega} v^2 - \int_{\partial \Omega} \frac{1}{2} \vec{\beta} \cdot \vec{n} v^2} + \gamma \|v\|_{L^2}^2$$

$$\begin{aligned} & \geq \underbrace{\min(\alpha, \gamma)}_{\geq 0} (\underbrace{\|\nabla v\|_{L^2}^2 + \|v\|_{L^2}^2}_{\|v\|_{H^1}^2}) - \underbrace{\frac{1}{2} \int_{\partial \Omega} \vec{\beta} \cdot \vec{n} v^2}_{\geq 0 \text{ if } \vec{\beta} \cdot \vec{n} \leq 0 \text{ on } \partial \Omega} \\ & \geq 0 \end{aligned}$$

- linear functional bounded.

$$\left| \int_{\Omega} f v \right| \leq \|f\|_{L^2} \|v\|_{L^2} \leq \|f\|_{L^2} \|v\|_{H^1} \quad \text{if } f \in L^2(\Omega).$$

Conclusion:

if $f \in L^2(\Omega)$ and $\beta \cdot n \leq 0$ on $\partial\Omega$

Then existence and unique solution.

2.3 (2pt):

$$V_h = \{v_h \in C^0(\Omega) : v_h|_K \in P^2(K) \quad \forall K \in \mathcal{T}_h\}$$

$$\text{Find } u_h \in V_h : a(u_h, v_h) = \int_{\Omega} f v_h \quad \forall v_h \in V_h.$$

$$\text{Stability: } \|u_h\|_{H^1(\Omega)} \leq \frac{1}{\min(\alpha, \gamma)} \|f\|_{L^2(\Omega)}$$

convergence:

$$\|u - u_h\|_{H^1(\Omega)} \leq C h^2 |u|_{H^3(\Omega)}$$

2.4 (1pt):

$$Lu = -\operatorname{div}(\alpha \nabla u) - \beta \cdot \nabla u + \gamma u = f$$

iteration n:



$$\begin{cases} Lu_1^{n+1} = f & \text{in } \Omega_1 \\ u_1^{n+1} = \theta u_2^n + (1-\theta) u_1^n & \text{on } \Gamma \\ \alpha \frac{\partial u_1^{n+1}}{\partial n} = 0 & \text{on } \partial\Omega_1 \setminus \Gamma \end{cases}$$

$$\begin{cases} Lu_2^{n+1} = f & \text{in } \Omega_2 \\ \alpha \frac{\partial u_2^{n+1}}{\partial n} = \alpha \frac{\partial u_1^{n+1}}{\partial n} & \text{on } \Gamma \\ \alpha \frac{\partial u_2^{n+1}}{\partial n} = 0 & \text{on } \partial\Omega_2 \setminus \Gamma \end{cases}$$

If $0 < \theta < \theta_{\max}$, for a suitable $\theta_{\max}$, the DD method converges. If $\theta = 1$ (no relaxation) the method does not necessarily converge.